

SMOKE ALARM

Made With UM3562

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This smoke alarm made using UM3562 sound generator can help maintain smoke-free environment in apartments, offices, factories, restaurants, hospitals, etc. In this article, we'll explore the workings of this circuit and how to build it.

Fig. 1 shows the author's prototype while Fig. 2 shows the MQ2 sensor module used for the smoke alarm. Circuit diagram for the alarm is shown in Fig. 3.

The circuit comprises a step-down transformer (X1), bridge rectifier (BR1), 5V voltage regulator LM7805 (IC1), MQ2 sensor module (connected across CON2), op-amp LM358 (IC2), sound generator UM3562 (IC3), three transistors (T1 through T3), an 8-ohm 0.5-watt loudspeaker (LS1), and several other components. Capacitor C1, connected across the supply terminals, minimises voltage ripples and noise signals.

The circuit works on 5V DC, which it derives from the transformer X1. The 9V AC secondary of X1 connects to bridge rectifier BR1 for rectification, which then feeds the regulator LM7805 to obtain a stable 5V supply for the circuit. The glowing LED1 indicates the presence of smoke near the sensor module.

The heart of the circuit lies in the MQ2 smoke sensor module and the sound generator IC UM3562. The smoke sensor module used here has three pins, but you can opt for a four-pin module as well.

The output of the MQ2 sensor module is connected to pin 3 of IC2,

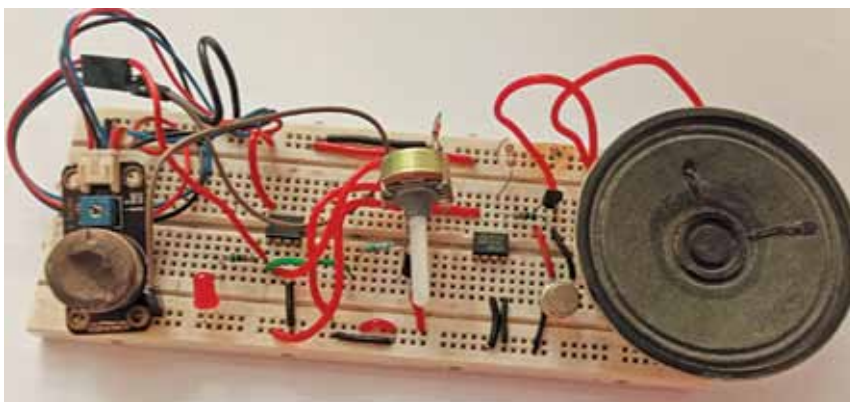


Fig. 1: Author's prototype

while pin 2 is linked to potentiometer VR1 for sensitivity control. IC2 serves as a comparator in the circuit. The output of IC2 connects to the base of transistor T1 via resistor R2 and also to the anode of LED1 through resistor R3.

The sound generator IC UM3562 (IC3) is a low-power CMOS LSI chip in an 8-pin dual in-line (DIP) package. It includes a voltage-controlled oscillator (VCO), timer generator, tri-state selector switch, envelope control, and mixer. Pin 2 of IC3, along with trigger pin 4, functions as a tri-state selector



Fig. 2: MQ2 sensor module

PARTS LIST

Semiconductors:

IC1	- LM7805, 5V voltage regulator
IC2 (ICA, IC2C)	- LM358 op-amp
IC3	- UM3562 sound generator
T1, T3	- 2N2219 NPN transistor
T2	- BC547 NPN transistor
BR1	- 1A bridge rectifier
LED1	- 5mm LED
ZD1	- 3.1V zener diode

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 220-ohm
R2, R3, R4, R6	- 1-kilo-ohm
R5	- 68-kilo-ohm
VR1	- 100-kilo-ohm pot

Capacitors:

C1	- 1000 μ F, 35V electrolytic
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Miscellaneous:

CON1	- 2-pin connector
CON2	- 3-pin connector
LS1	- 8-ohm, 0.5-watt speaker
X1	- 230V AC primary to 9V, 500mA secondary transformer
	- MQ2 sensor module

switch, producing three types of sounds. In this design, we have not connected pin 2, so it is in a floating condition, generating the machine-gun sound.

IC3 operates on a 3V battery, so the 5V is reduced to the required